1. Polynomial Models

```
linModel<-lm(testscr~avginc)
summary(linModel, robust=T)

plot(testscr~avginc, caschool)
abline(linModel, col="red")

hist(caschool$avginc)

# Creating New Variables

caschool$avgincsq=caschool$avginc^2

#Alternatively,
quadModel<-lm(testscr~avginc+I(avg^2), caschool)
summary(quadModel,robust=T)

anova(quadModel,linModel)


caschool$avginccu=caschool$avginc^3
CubicModel<-lm(testscr~avginc+avgincsq+avginccu, caschool)
summary(CubicModel,robust=T)

anova(CubicModel,linModel )
```

Logarithms

Case	Population regression function	Interpretation of Coefficient
1. linear-linear	$Y_i = \beta_0 + \beta_1 X_i + u_i$	$\Delta Y = \beta_1 \Delta x$
2. linear-log	$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$	$\Delta Y = (\beta_1/100)\%\Delta x$
3. log-linear	$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$	$\%\Delta Y = 100\beta_1 * \Delta x$
4. log-log	$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i$	$\%\Delta Y = \beta_1 \%\Delta x$

```
linlin<-lm(testscr~avginc, caschool)
summary(linlin,robust=T)

linlog<-lm(testscr~I(log(avginc)), caschool)
summary(linlog,robust=T)

loglin<-lm(I(log(testscr))~avginc, caschool)
summary(loglin,robust=T)

loglog<-lm(I(log(testscr))~I(log(avginc)), caschool)
summary(loglog,robust=T)
```

Interaction Variables

```
wagemodel<-lm(wage~female+exper, wageData)
summary(wagemodel, robust = T)

wagemodel<-lm(wage~female+exper+I(exper*female), wageData)
summary(wagemodel, robust = T)
```